Automatic Registration of Point Clouds with Building Models

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Keywords: urban scene, ICP registration, pre-alignment, weighted ICP

Abstract

In urban scenes, aligning real scanned point cloud data with building models is a fundamental step for many applications. However, under real data conditions, achieving stable and automatic registration still remains challenging.
This study focuses on the registration problem between TLS point clouds and CityGML building models in real urban scenes, and develops a multi-stage ICP-based registration method. The experimental method tackles challenges such as partial overlap, the presence of surroundings, structural similarity, and incomplete data through a pipeline including pre-alignment, spatial cropping, candidate generation, and weighted ICP.
The experimental results show that, at the building level, the pipeline can achieve stable and accurate registration performance, as reflected by a reduction in RMSE. At the building-group level, the registration result remains visually reasonable, but due to the limited overlap, the fitness metric shows certain limitations.
These results indicate that the proposed pipeline is effective under building-level scenarios, while evaluation metrics require further consideration under low-overlap conditions.
Code availability: The implementation of the proposed method is available at `https://gitlab.lrz.de/00000000014BE35A/automatic-registration-of-point-clouds-with-building-models.git`.

1. Introduction

In urban scenes, aligning real scanned point cloud data with building models is a fundamental step for many applications. However, under real data conditions, achieving stable and automatic registration remains challenging (Xu et al., 2024).

This study focuses on building-level registration, aligning terrestrial laser scanning (TLS) point clouds with CityGML building models. In this problem, the two types of data show clear differences: TLS point clouds usually only cover part of the building facades, contain street-level clutter and occlusions, and also exhibit uneven point density and incomplete scanning; while CityGML models provide complete and regular building geometry, but lack real observation details. This difference makes it difficult to directly establish correspondences between the two datasets.

In practical scenarios, the following problems mainly exist: first, the overlap between TLS and the model is small; second, there are multiple structurally similar facades in urban scenes, which can easily lead to incorrect matching; in addition, non-building structures and point density imbalance in the scanned data can also interfere with the registration process. In such cases, directly applying ICP often leads to convergence to incorrect positions (Goebbels et al., 2019; Yang et al., 2016), aligning wrongly to the ground or to other similar facades, as illustrated in Figure 1, where the alignment is attracted to the ground or adjacent facades instead of the correct building surface.

To address these problems, this work constructs a multi-stage registration pipeline based on ICP. Before the actual registration, the data are constrained through pre-alignment and spatial range restriction, in order to reduce the possibility of incorrect matching, and on this basis, weighted point-to-plane ICP is applied to progressively complete the registration. The pipeline is extended from a single wall to a single building and further to a building group, in order to analyze its performance and limitations.

2. Method

2.1 Overview

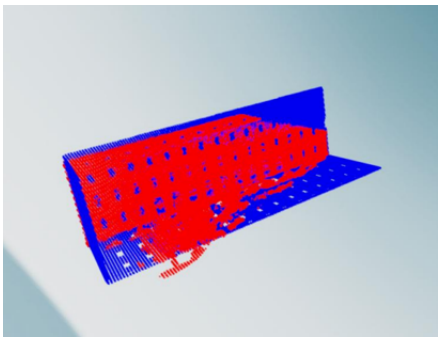
The overall pipeline adopted in this study mainly includes pre-alignment, spatial range restriction, wall-based candidate generation, and fine registration based on ICP, as illustrated in Figure 2. Directly aligning the entire model makes it difficult to obtain effective registration results; therefore, this study first constrains the search space and then performs a more stable refinement.

The pipeline is developed progressively from simple scenarios to more complex ones. It starts from simple single planar cases, where point-to-plane ICP, which is less sensitive to noise, is used for facade alignment; then additional constraints such as pre-alignment and weighted ICP are gradually introduced. For building-level cases, spatial range restriction and candidate-based strategies are further introduced to handle ambiguity caused by structural similarity, especially the confusion between similar facades within the same building group.

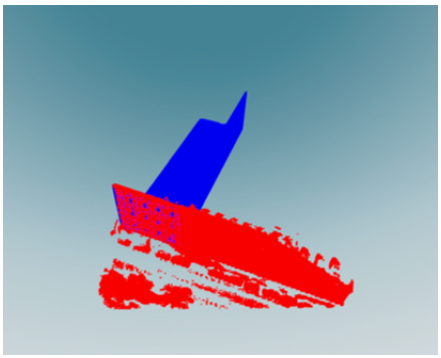
Among these components, cropping plays a central role. By restricting the spatial region before registration, it guides the alignment process to focus more on the correct area and reduces the risk of being attracted to symmetric or unrelated structures. Based on this, the subsequent ICP refinement process becomes more stable.

2.2 Pre-alignment

Under real data conditions, initial position errors have a significant impact on the registration result. If the initial alignment



(a) Incorrect alignment to ground



(b) Incorrect alignment to adjacent facade

Figure 1. Typical incorrect convergence of ICP in urban scenes. Due to limited overlap and structural ambiguity, the TLS point cloud (red) can be incorrectly aligned to the ground (a) or to an adjacent facade (b) of the CityGML model (blue).

is not reasonable, ICP can easily converge to incorrect positions, such as aligning walls to the ground or to adjacent or opposite facades. Therefore, before performing fine registration, an initial alignment is required, as illustrated in Figure 3, where the datasets are significantly misaligned before initialization and become roughly aligned after coarse transformation.

In this study, a geometry-constrained pre-alignment method is adopted. First, in the Z direction, lower quantile points in the point cloud are selected as a ground approximation to align the two datasets, in order to reduce vertical deviation. In the XY plane, centroid alignment is used to provide an initial translation. At the same time, a yaw constraint is introduced in the transformation, allowing only rotation around the vertical axis, so as to avoid unreasonable orientation changes.

In addition, in order to reduce the influence of interfering structures, normal-based filtering is applied to

the point cloud, and only approximately vertical structures are retained for the subsequent registration process. This step serves as a weak semantic constraint and helps to improve the robustness of the initial alignment.

Through the above pre-alignment steps, obvious mismatches can be avoided to some extent, providing a larger initial overlap between the target data and the source data, ensuring that ICP can converge normally.

2.3 Cropping

In building-level registration problems, TLS point clouds usually only cover part of the outer facades of a building, while CityGML models contain complete structures. This leads to a small effective overlap between the two (Chen et al., 2020), while the model also contains a large amount of surroundings that are not related to the building itself. These surroundings can easily introduce incorrect correspondences during the registration process, thereby affecting the stability of ICP, as illustrated in Figure 4.

To address this issue, this study introduces a spatial range-based cropping operation after pre-alignment. Specifically, taking the transformed source data as reference, an axis-aligned bounding box (AABB) is constructed, and padding is added to its minimum and maximum bounds to crop the target model, so that only the regions relevant to the current observation are retained. Similar spatial constraint strategies have also been applied in multi-stage registration methods (Hou et al., 2025).

Through this step, the influence of irrelevant structures on the registration process can be effectively reduced and the actual overlap between the source and target data can be improved. In addition, cropping also alleviates the ambiguity caused by structural similarity to some extent, reducing the risk that the registration result is attracted to symmetric or incorrect facades.

2.4 Wall-based Candidate Selection

In building-scale scenarios, structurally similar facades introduce matching ambiguity. To address this issue, this study introduces a wall-based candidate generation strategy on the cropped target data. First, normal estimation is performed on the target point cloud, and approximately vertical structures are selected to remove the influence of ground and other non-building structures. Based on this, these normals are clustered into several main directions, thereby obtaining different candidate regions for walls.

Subsequently, the source data is registered with each candidate region respectively, and the registration results are compared based on evaluation metrics, from which the optimal alignment result is selected as the final output (Theiler and Schindler, 2012). This strategy explicitly generates multiple candidates and evaluates them, which to some extent avoids ICP directly converging to incorrect facades, and can effectively reduce the matching ambiguity caused by structural similarity.

2.5 Weighted ICP

After spatial range restriction and candidate selection, point-to-plane ICP is applied to perform fine registration between the source and target data. Its basic formulation can be written as:

$$\min_{T} \sum_{i} \left(n_i^T \left(p_i - T(q_i) \right) \right)^2, \tag{1}$$

where the optimal transformation is obtained by minimizing the point-to-plane distances.

In real scanned data, due to uneven point density, surroundings, and unreliable correspondences, directly applying standard ICP is highly sensitive to local regions, which reduces the stability of the registration. Therefore, based on the above naive ICP, a weighted formulation is introduced:

$$\min_{T} \sum_{i} w_i \left(n_i^T \left(p_i - T(q_i) \right) \right)^2, \tag{2}$$

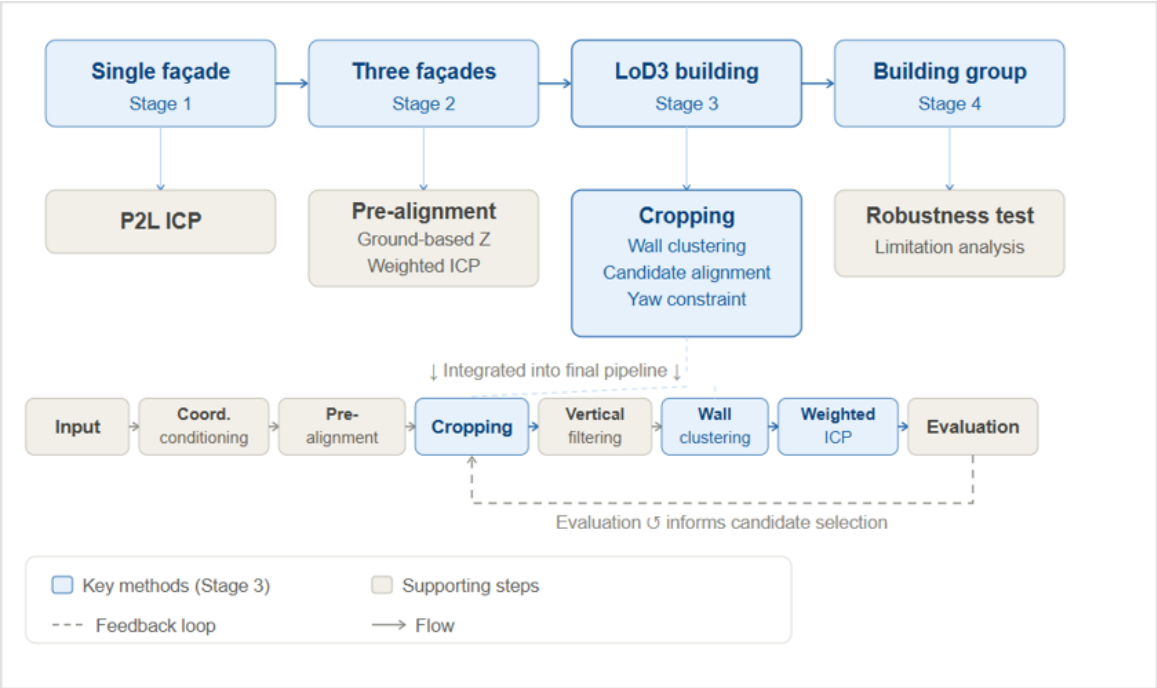
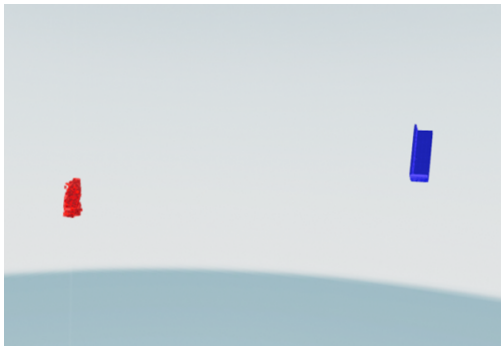
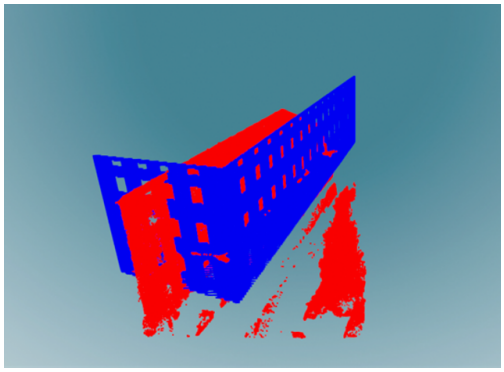


Figure 2. Overview of the proposed multi-stage registration pipeline. The workflow includes pre-alignment, spatial cropping, wall-based candidate selection, and weighted ICP refinement across progressively complex scenarios.

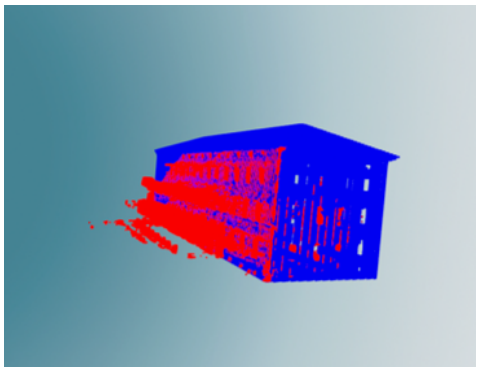


(a) Before pre-alignment

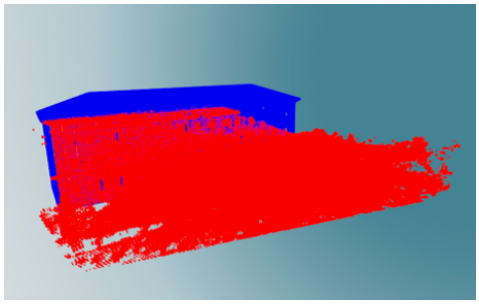


(b) After pre-alignment

Figure 3. Effect of the pre-alignment step. Before initialization (top), the TLS point cloud (red) and the CityGML model (blue) are significantly misaligned. After applying coarse alignment (bottom), the two datasets are roughly aligned, providing a reliable starting point for subsequent ICP refinement.



(a) Without cropping



(b) With spatial cropping

Figure 4. Effect of spatial cropping in resolving structural ambiguity. Without proper spatial constraints (top), the registration is attracted to a symmetric but incorrect facade. After applying cropping (bottom), the search space is restricted and the correct facade is successfully selected.

where the weight w_i is used to adjust the influence of different correspondences in the optimization. In this study, the weights mainly consider two aspects: first, density weighting based on voxel statistics is used to reduce the dominance of high-density regions; second, a robust kernel is applied to the residuals to reduce the influence of outliers and incorrect correspondences (Cao et al., 2023; Li et al., 2020; Liu et al., 2018).

In addition, during correspondence establishment, a maximum correspondence distance and distance-based filtering are applied to reject unreasonable matches, thereby reducing the influence of incorrect correspondences.

To further improve convergence stability, a multi-stage coarse-to-fine threshold strategy is adopted in the ICP process. Starting from a larger search range, the correspondence distance is gradually reduced, resulting in a more stable fine registration process.

Through the above weighting and constraint strategies, the influence of density imbalance, outliers, and unreliable correspondences on the registration result can be reduced to some extent, thereby improving the overall registration stability.

3. Experiments

3.1 Experimental Setup

This experiment uses two types of data: TLS point clouds and point clouds generated from CityGML building models, as illustrated in Figure 5, including their spatial distribution and the corresponding urban context.

The TLS point cloud is acquired by mobile laser scanning, with the acquisition area located on the entrance side of the TUM main campus. It mainly covers the facades on the entrance side of the TUM campus, while also including the street environment in front of the facades, such as trees, street space, and building facades across the street. This dataset only contains partial wall observations and exhibits uneven point density and incomplete coverage. The CityGML building model describes the complete structure of the TUM campus building group, including multiple buildings and their regular planar walls, and is provided in the UTM coordinate system.

In the experiment, both datasets are first preprocessed in a unified way, including voxel downsampling and normal estimation. Then, following the method pipeline described in the previous section, pre-alignment, spatial cropping, candidate generation, and weighted multi-stage ICP fine registration are performed. The main parameters include the voxel size and the correspondence distance thresholds at different ICP stages. The registration results are evaluated using quantitative metrics, including root mean square error (RMSE) and fitness.

3.1.1 Single-wall Case In simple single-wall scenarios, the registration problem is relatively clear, and there is good geometric consistency between the source and target data. In this case, the naive ICP algorithm can converge relatively stably and quickly and obtain reasonable and reliable registration results. Compared with point-to-point ICP, point-to-plane ICP performs more stably on planar structures and converges faster, which is suitable for this experimental scenario. Therefore, in the subsequent experiments under more complex scenarios, this study continues to adopt point-to-plane ICP as the basic registration framework.

3.2 Results and Analysis

3.2.1 Multiple-wall Cases In multiple-wall cases, due to differences in spatial position and orientation between different walls, initial position errors have a significant impact on the convergence of ICP, leading to unstable results or even failure to achieve effective registration.

After introducing pre-alignment, by performing initialization in the Z direction based on a ground approximation, and combining translation alignment in the XY plane with a yaw constraint, a more reasonable initial position can be provided for subsequent registration. As shown in Figure 6, standard ICP tends to converge to incorrect planar structures, while the proposed weighted ICP significantly improves the alignment accuracy.

These results indicate that, in multi-wall scenarios, proper initialization plays an important role in improving the stability of ICP. The convergence process becomes more stable, and incorrect alignment to the ground or other unrelated structures is significantly reduced.

In building-level scenarios, the TLS data only covers part of the facade, while the CityGML model contains the complete building structure, and structurally similar wall regions also exist, making it more difficult to establish correct correspondences.

In this case, the complete pipeline consisting of cropping, candidate generation, and weighted ICP works together to achieve a stable registration result. The source and target data are well aligned at the building level, and the correct facade is successfully selected.

To further evaluate the registration performance, quantitative metrics are introduced in addition to visual inspection. The results show that the multi-stage ICP strategy significantly improves convergence, reducing the RMSE from about 1.55 to 0.17. At the same time, using a smaller voxel size further improves the performance, as reflected by increasing fitness and decreasing RMSE. The resulting alignment is geometrically consistent and accurate at the building scale.

It should be noted that fitness is more sensitive to the coverage of the overlapping region, while RMSE mainly reflects the geometric error of the matched points, and therefore their trends are not strictly consistent. With a smaller voxel size, more points are involved in matching, leading to a significant increase in fitness, while the newly included points may contain edge and noisy regions, resulting in a relatively limited improvement in RMSE.

Through the cluster-based candidate evaluation strategy, incorrect convergence to symmetric or wrong facades can be effectively avoided, enabling more reliable alignment in scenarios with structural similarity.

Table 1. Quantitative results for building-level registration under different voxel sizes.

Voxel size	Fitness ↑	RMSE ↓
1.0 m	0.2207	0.4309
0.5 m	0.4616	0.3473
0.3 m	0.6040	0.3321

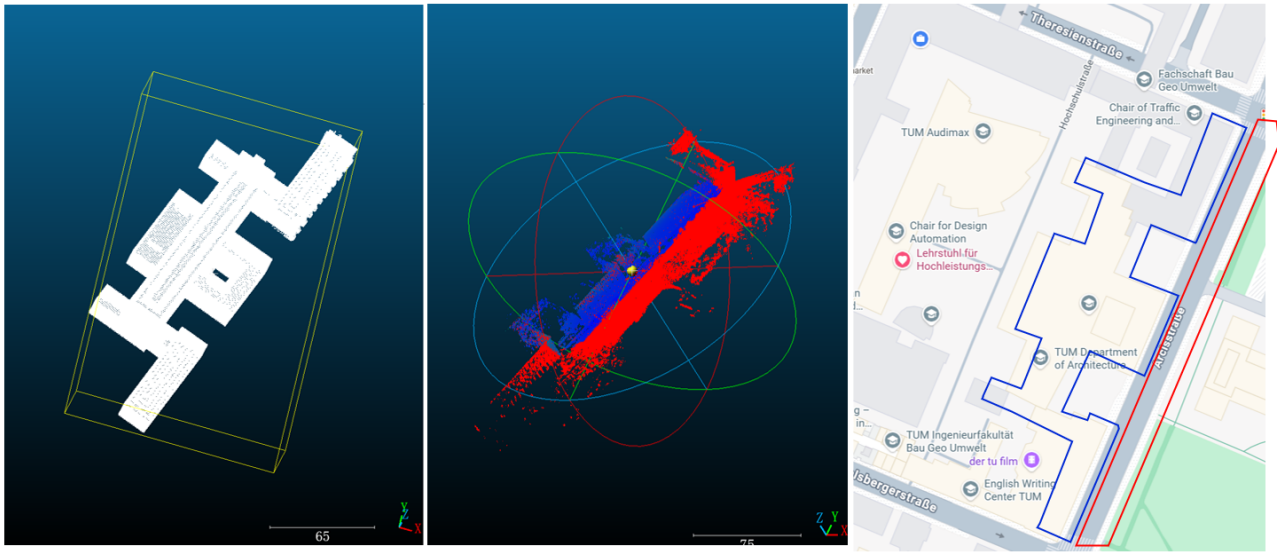
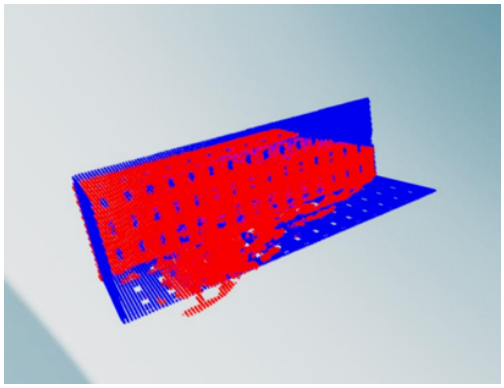
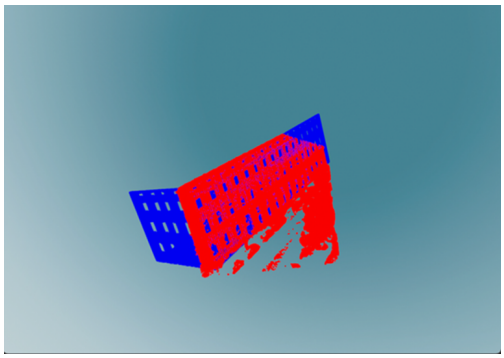


Figure 5. Example data used in this study. The TLS point cloud (red) captures partial facade observations with surrounding clutter, while the CityGML model (blue) provides complete building geometry. The right panel shows the corresponding spatial context of the study area.



(a) Standard ICP result



(b) Weighted ICP result

Figure 6. Comparison of registration results in multi-wall scenarios. Standard ICP (top) tends to be attracted to incorrect planar structures due to structural ambiguity, while the proposed weighted ICP (bottom) achieves more accurate alignment by reducing the influence of density imbalance and incorrect correspondences.

3.2.2 Building-group Case and Limitations In building-group scale registration scenarios, the TLS data only covers partial facades, while the CityGML model contains the complete structure of multiple buildings, which further reduces the actual overlap between the two. In this case, using the same registration pipeline as described above, the registration result appears visually reasonable at the building-group scale, even though the quantitative metrics indicate limited overlap, as shown in Figure 7.

The fitness values are generally low, while RMSE remains within a relatively stable range. This is mainly because a large number of model points have no corresponding observations in the TLS data, resulting in a low proportion of valid correspondences. Therefore, low fitness in this scenario reflects insufficient overlap rather than poor alignment quality.

At the same time, it can be observed that as the voxel size decreases, the fitness still shows an increasing trend, while RMSE decreases slightly. This indicates that even under low-overlap conditions, the quantitative metrics can still reflect the relative changes in registration performance.

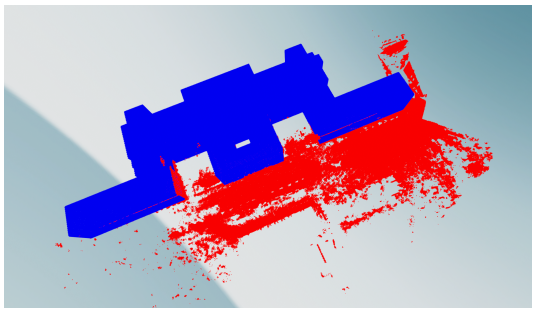
These results indicate that, on the building-group scale, the traditional evaluation metric based on the proportion of correspondences (fitness) has certain limitations. In future work, more suitable evaluation metrics can be considered for low-overlap scenarios to more accurately reflect registration quality.

Table 2. Quantitative results for building-group registration under different voxel sizes.

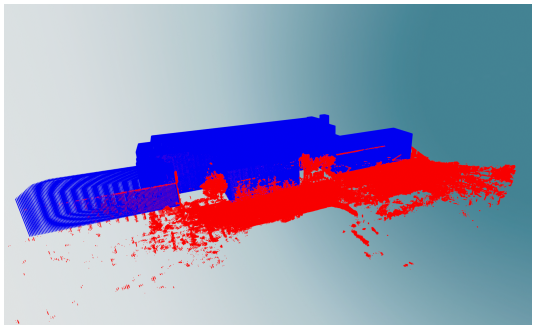
Voxel size	Fitness ↑	RMSE ↓
1.0 m	0.0317	0.4604
0.5 m	0.0444	0.4301
0.3 m	0.0596	0.4230

4. Discussion and Conclusion

This study focuses on the registration problem between TLS point clouds and CityGML building models in real urban scenes



(a) Top view



(b) Side view

Figure 7. Registration result in the building-group scenario visualised from two different viewpoints. The top view (top) highlights the overall spatial alignment between the TLS point cloud (red) and the CityGML model (blue), while the side view (bottom) reveals the vertical consistency and remaining discrepancies.

and develops a multi-stage ICP-based registration method. The experimental method tackles challenges such as surroundings, structural similarity, and incomplete data through a pipeline including pre-alignment, spatial cropping, candidate generation, and weighted ICP.

The experimental results show that, at the building level, the pipeline can achieve stable and accurate registration performance. By introducing density-based weighting, the convergence performance is significantly improved, as reflected by a reduction in RMSE. Meanwhile, cropping and the candidate-based selection strategy are effective in resolving ambiguity caused by structural symmetry. At the building-group level, the registration result remains visually reasonable, but due to the limited overlap, the fitness metric shows certain limitations; therefore, more appropriate evaluation metrics are still needed for analysis. Overall, the proposed multi-stage framework provides a robust and interpretable solution for point cloud to building model registration under realistic urban conditions.

Regarding the method, the window-based prior is not effective enough in this experiment so far, as the stability decreases significantly with increasing cropping range. The current method still has some dependence on initialization and parameter settings.

Future research directions can be explored as follows: first, scaling up to more diverse building datasets and more challenging data to validate the generalization ability of the current method; second, exploring more automatic initialization strategies to reduce dependence on prior constraints; in addition, further analyzing the impact of density, incompleteness, and surroundings on registration performance, and comparing rigid versus non-rigid registration strategies; finally, considering the introduction of non-ICP methods or more data-driven methods. These directions will contribute to more reliable and scalable registration in complex urban environments.

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